B1 position-aware statute queries: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Ask each statute label to attend to document chunks while preserving their order. This replaces a uniform document readout with label-conditioned chunk selection, testing whether different statutes need different parts of the judgment.

Architecture and mechanism

B1 produces four window representations. Learned statute queries and position-aware cross-attention gather evidence before multi-label classification.

Measured result

Scored submission	1016786
Codabench macro-F1	0.2254
Archived local metric	0.2299 official-dev macro-F1

Local official-dev macro-F1 is about 0.2299 and Codabench is 0.2254. The result does not support an improvement over simpler LoRA or the full-text hybrid in this run.

Limitations and next check

A query can only attend to the four represented windows, so adding attention does not remove the upstream context bottleneck. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase5_position_queries.py, phase5_multiwindow.py, phase5_b1_train.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.